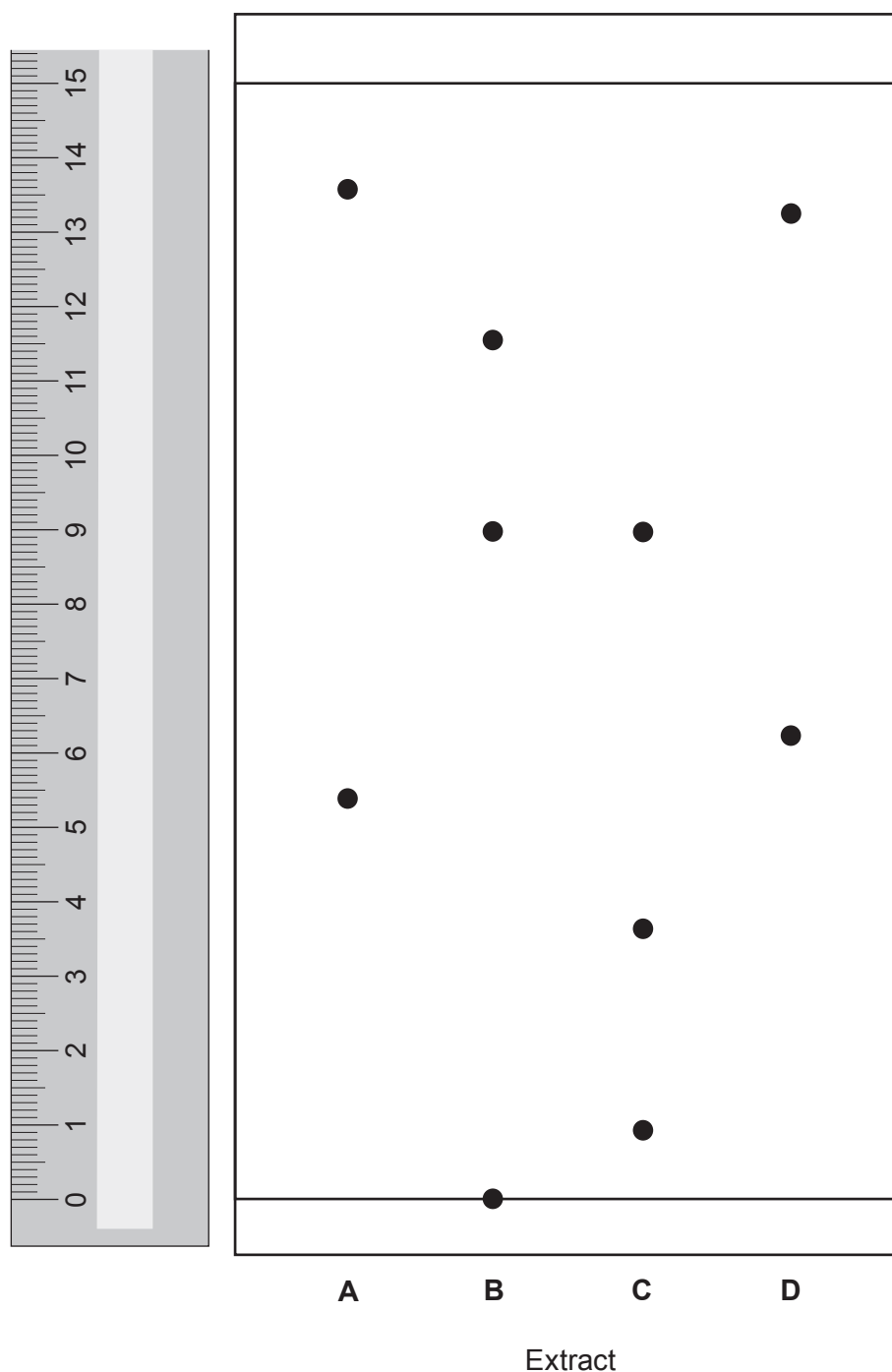


2. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



- (a) All of the extracts contain a mixture of pigments with different  $R_f$  values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest**  $R_f$  value 0.60?

Give your reasoning.

[3]

Extract .....

Reasoning .....

.....  
 .....

- (b) Explain why the pigments travel different distances on the chromatogram.

[2]

.....  
 .....  
 .....

- (c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice.

[2]

Extract .....

Explanation .....

.....

- (d) The chemical formula of the solvent acetone is  $C_3H_6O$ . Calculate the percentage by mass of carbon in acetone.

The relative formula mass ( $M_r$ ) of acetone is 58.

[2]

$$A_r(C) = 12$$

Percentage = ..... %

